SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage: 40%

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Real-Time System Design Trade-off

A surveillance system must operate under constraints:

- Viola-Jones: fast but less accurate (78%)
- YOLO: moderate speed, good accuracy (92%)
- Deep Learning: high accuracy (95%) but slow

Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

2. Robust Detection under Environmental Variations

Detection accuracy across environments:

Method	**Normal**	**Occlusion**	**Low Light**

Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Compare the robustness of each method and evaluate which technique provides the best overall performance across varying conditions. Justify trade-offs.

3. Optical Flow Reliability vs Noise Sensitivity

Two optical flow methods show:

- Method A: stable vectors but misses small motions
- Method B: detects fine motion but noisy vectors

Compare both methods and evaluate which is more suitable for motion tracking in dynamic scenes. Justify based on application requirements.

4. Background Modeling Strategy (GMM)

A GMM-based system shows:

- High false positives in dynamic background
- Missed detections in static background

Compare GMM performance across scenarios and evaluate whether GMM alone is sufficient. Propose and justify improvements.

5. Motion Estimation vs Computational Cost

Two methods:

- Method A: low error (5%), high computation
- Method B: moderate error (10%), low computation

Compare the methods and evaluate their suitability for real-time vs offline applications. Justify your decision.

6. Structure from Motion Consistency

Two datasets produce:

- Dataset A: accurate depth but requires high-quality images
- Dataset B: inconsistent depth but works on low-quality images

Compare the datasets and evaluate which setup is more practical for real-world deployment. Justify your reasoning.

7. Stereo Vision vs Motion Estimation

Depth estimation results:

- Stereo Vision: accurate but requires dual cameras
- Motion Estimation: less accurate but works with single camera

Compare both approaches and evaluate which is preferable under hardware constraints. Justify your choice.

8. Model Generalization vs Overfitting

Performance:

Model	**Training**	**Testing**
A	98%	70%
B	90%	88%

Compare the models and evaluate which is more reliable for deployment. Justify based on generalization and risk factors.

9. YOLO Performance across Scales

Detection results:

- Large objects: 95%
- Medium objects: 85%
- Small objects: 60%

Compare performance across scales and evaluate whether YOLO alone is sufficient for multi-scale detection tasks. Justify improvements.

10. Motion Model Selection for Animation

Two models:

- Model A: smooth but physically inaccurate
- Model B: realistic but computationally intensive

Compare the models and evaluate which should be used for animation systems under different constraints (real-time vs cinematic). Justify your decision.

11. Detection Model Comparison under Resource Constraints

Performance:

Model	Accuracy	Latency
A	85%	40 ms
B	92%	90 ms
C	90%	60 ms

Compare the models and evaluate which is optimal for real-time systems with moderate accuracy requirements. Justify your decision.

12. Robustness vs Speed Trade-off

Performance:

Method	Accuracy	Robustness	Speed
A	88%	High	Slow
B	85%	Moderate	Fast
C	90%	Low	Moderate

Compare the methods and evaluate the best choice for a balanced system. Justify trade-offs.

13. Optical Flow Stability Analysis

Results:

Method	Accuracy	Noise Sensitivity
A	High	High
B	Moderate	Low

Compare both methods and evaluate suitability for real-time tracking in noisy environments.

14. Motion Estimation Accuracy vs Efficiency

Results:

Method	Error	Computation
A	4%	High
B	9%	Low
C	6%	Moderate

Compare methods and evaluate the best approach for embedded systems.

15. GMM Performance in Different Conditions

Results:

Scenario	Accuracy
Static	92%
Dynamic	68%

Compare performance and evaluate whether GMM is reliable for all environments. Suggest improvements.

16. Object Detection under Scale Variation

Results:

Object Size	Accuracy
Large	94%
Medium	80%
Small	55%

Compare detection performance and evaluate system limitations. Justify improvements.

17. Stereo vs SfM Accuracy Comparison

Results:

Method	Accuracy	Hardware
Stereo	High	Dual camera
SfM	Moderate	Single camera

Compare and evaluate which is more practical under hardware limitations.

18. Detection Model Generalization

Results:

Model	Training	Testing
A	97%	75%
B	90%	88%

Compare and evaluate which model is better for deployment.

19. Motion Tracking Accuracy Comparison

Results:

Method	Accuracy	Stability
A	High	Low
B	Moderate	High

Compare and evaluate which is suitable for long-term tracking.

20. Surveillance Model Selection

Results:

Model	Accuracy	Latency
A	82%	30 ms
B	90%	70 ms
C	95%	120 ms

Compare and evaluate under strict latency constraints.

21. Noise Impact on Detection

Results:

Noise Level	Accuracy
Low	92%
Medium	80%
High	60%

Compare performance and evaluate system robustness.

22. Depth Estimation Error Analysis

Results:

Method	Error
A	0.5 m
B	1.2 m
C	0.8 m

Compare and evaluate accuracy for real-world applications.

23. Multi-Model Comparison

Results:

odel	Accuracy	Speed	Robustness
A	88%	Fast	Low
B	92%	Moderate	High
C	95%	Slow	High

Compare and evaluate the best overall model.

24. Motion Detection under Crowd Density

Results:

Method	Sparse Crowd	Dense Crowd
Optical Flow	High	Low
Motion Models	Moderate	High

Compare and evaluate best approach.

25. Model Efficiency vs Accuracy

Results:

Model	Accuracy	Computation
A	90%	High
B	85%	Low
C	88%	Moderate

Compare and evaluate for edge devices.

26. Feature Detection under Illumination Changes

Results:

Method	Normal	Low Light
A	90%	60%
B	85%	75%

Compare and evaluate robustness.

27. Video Processing Pipeline Comparison

Results:

Pipeline	Accuracy	Speed
A	88%	Fast
B	92%	Moderate
C	95%	Slow

Compare and evaluate suitability.

28. Motion Model Evaluation

Results:

Model	Realism	Speed

A	High	Low
B	Moderate	High

Compare and evaluate based on application context.

29. Detection under Occlusion

Results:

Method	Accuracy
A	70%
B	85%
C	90%

Compare and evaluate robustness under occlusion.

30. System Performance under Constraints

Results:

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Compare and evaluate the best system considering all constraints.

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

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Course Name: Computer Vision   Course Code: ITA05

VIGNESHWAR S V

REGNO: 192321135

12. Robustness vs Speed Trade-off

Given Results

Method	Accuracy	Robustness	Speed
A	88%	High	Slow
B	85%	Moderate	Fast
C	90%	Low	Moderate

Comparison

Method A

- Accuracy is 88%.
- It has high robustness, meaning it performs reliably under different conditions such as noise, illumination changes, or environmental variations.
- However, it is slow and requires more computational time.
- Suitable where reliability is more important than processing speed.

Method B

- Accuracy is only 85%, the lowest among the three.
- It has moderate robustness.
- Its major advantage is high speed.
- Suitable for real-time applications where fast response is essential.

Method C

- It provides the highest accuracy of 90%.
- However, its robustness is low, so performance may decrease significantly when operating conditions change.
- Its speed is moderate.
- Suitable for controlled environments where accuracy is the primary requirement.

Best Choice for a Balanced System

For a balanced system, Method A is the best choice.

Although Method C gives the highest accuracy, its low robustness makes it unreliable in changing real-world conditions. Method B is very fast but has lower accuracy and only moderate robustness.

Method A provides a good combination of:

- High accuracy – 88%
- High robustness
- Acceptable performance for applications where reliability is important

Trade-off

The choice depends on the application:

Requirement	Best Method
Highest accuracy	C
Highest speed	B

Highest robustness	A
Balanced reliability	A
Real-time processing	B
Controlled environment	C

Conclusion

Method A is recommended for a balanced system because robustness is important in real-world applications. The additional computational cost is justified when consistent performance is required. If strict real-time processing is required, Method B may be preferred, while Method C is suitable when maximum accuracy is more important than robustness.

10. Motion Model Selection for Animation

Given Models

- Model A: Smooth but physically inaccurate
- Model B: Realistic but computationally intensive

Model A – Smooth Motion

Advantages:

- Produces visually smooth animation.
- Requires relatively low computational resources.
- Suitable for real-time rendering.
- Can maintain high frame rates.
- Useful for interactive applications.

Disadvantages:

- Motion may not obey real-world physical laws.
- Objects may appear unrealistic.
- Less suitable for simulations requiring physical accuracy.

Model B – Realistic Motion

Advantages:

- Produces physically realistic movement.
- Better representation of gravity, acceleration, collision, and object dynamics.
- Suitable for high-quality animation and simulation.

Disadvantages:

- Requires more computational power.
- Rendering and simulation can be slower.
- May not be suitable for systems requiring immediate response.

Comparison

Feature	Model A	Model B
Smoothness	High	High
Physical accuracy	Low	High
Computational cost	Low	High
Real-time suitability	Excellent	Limited
Cinematic quality	Moderate	Excellent
Realism	Low	High

Decision Based on Application

Real-time applications:

Model A should be preferred because speed and smooth frame rates are important. Examples include:

- Video games
- Virtual reality
- Interactive applications

- Real-time visualization

Cinematic applications:

Model B should be preferred because realism is more important than processing speed. Examples include:

- Movies
- VFX
- Animation films
- High-quality simulations

Conclusion

There is a clear speed-versus-realism trade-off. Model A is appropriate when real-time performance is required, while Model B is preferable for cinematic systems where realistic motion is the priority.

22. Depth Estimation Error Analysis

Given Results

Method	Error
A	0.5 m
B	1.2 m
C	0.8 m

For depth estimation, lower error means better accuracy.

Comparison

Method A

- Error = 0.5 m
- Has the lowest error.
- Therefore, it provides the highest depth estimation accuracy.
- Suitable for applications requiring precise distance measurements.

Method B

- Error = 1.2 m
- Has the highest error.
- Therefore, it provides the lowest accuracy among the three.
- It may be unsuitable for applications where precise depth is critical.

Method C

- Error = 0.8 m
- Its performance is between A and B.
- Better than B but worse than A.

Ranking

A > C > B (where lower error represents better performance)

Rank	Method	Error	Accuracy
1	A	0.5 m	Highest
2	C	0.8 m	Moderate
3	B	1.2 m	Lowest

Real-World Applications

Depth estimation is important in:

- Autonomous vehicles
- Robotics
- 3D reconstruction
- Augmented reality
- Object distance measurement

- Navigation systems

For autonomous driving, for example, an error of 1.2 m could cause incorrect estimation of the distance between the vehicle and an obstacle.

Conclusion

Method A is the best choice because it has the lowest error of 0.5 m. Method C can be used when moderate accuracy is acceptable. Method B is least suitable for safety-critical applications because its error is the highest.

26. Feature Detection under Illumination Changes

Given Results

Method	Normal Light	Low Light
A	90%	60%
B	85%	75%

Method A

Under normal lighting: Accuracy = 90%

Under low light: Accuracy = 60%

Performance reduction: $90 - 60 = 30\%$

Therefore, Method A is highly affected by illumination changes.

Method B

Under normal lighting: Accuracy = 85%

Under low light: Accuracy = 75%

Performance reduction: $85 - 75 = 10\%$

Therefore, Method B is considerably more robust to illumination changes.

Comparison

Feature	Method A	Method B
Normal-light accuracy	90%	85%
Low-light accuracy	60%	75%
Accuracy drop	30%	10%
Robustness	Low	High

Evaluation

Method A has higher accuracy under normal conditions, but its performance drops significantly in low-light environments.

Method B has slightly lower normal-light accuracy but maintains much better performance under low-light conditions.

The important measure for robustness is the change in performance under different conditions.

Conclusion

Method B is the more robust feature detection method.

Although Method A achieves 90% accuracy in normal lighting, Method B maintains 75% accuracy in low light compared with A's 60%.

Therefore:

- For controlled normal lighting → Method A
- For varying illumination → Method B
- For real-world applications → Method B is preferable

28. Motion Model Evaluation

Given Results

Model	Realism	Speed
A	High	Low
B	Moderate	High

Model A

Model A provides high realism, meaning its generated motion closely represents real-world movement.

Advantages:

- Highly realistic animation.
- Suitable for simulations.
- Produces natural-looking motion.
- Useful when physical accuracy is important.

Disadvantage:

- Low processing speed.
- Requires more computational resources.
- May not be appropriate for real-time applications.

Model B

Model B provides moderate realism but has high speed.

Advantages:

- Fast computation.
- Suitable for real-time applications.
- Requires fewer computational resources.
- Can provide high frame rates.

Disadvantage:

- Motion is less realistic than Model A.

Comparison

Requirement	Model A	Model B
Realism	High	Moderate
Speed	Low	High
Computational requirement	High	Low
Real-time applications	Less suitable	Suitable
Cinematic applications	Suitable	Less suitable
Physical simulation	Suitable	Moderate

Application Context

Real-time systems:

Model B is preferable because speed is critical. Examples include:

- Games

- VR/AR
- Interactive simulations
- Real-time animation

Cinematic systems:

Model A is preferable because high realism is more important than speed. Examples include:

- Movies
- VFX
- Animation films
- High-quality rendered scenes

Conclusion

There is no universally best model. Model A should be selected when realism is the main requirement, while Model B should be selected when speed and real-time performance are more important.

Thus, the selection depends on the application's constraints and the required balance between realism and computational efficiency.